0.1 Pre-cursor: Methods

0.1.1 (smartly) projecting onto a 2-band subspace

In what follows we will encounter systems with more than two bands. In order to derive an effective theory for a small energy window around a Dirac-point most of these bands are irrelevant and can be safely projected out. Asssume a k-space Hamiltonian $H(\vec{q})$ and a degeneracy point (/Dirac point) $\vec{K}$. Furtheremore, I define

$$H(\vec{K}+\vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}}H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{1}$$

The next ingredient is a orthonormal basis of the degenerate subspace $\{\vec{v}_1, \vec{v}_2\}$. I also need the matrix elements of $H^1_{\vec{K}}$ wrt to this basis:

$$\begin{aligned}
\vec{v}_1^H \cdot H^1_{\vec{K}} \cdot \vec{v}_1 &\equiv M_{11} = M^x_{11}p_x + M^y_{11}p_y \\
\vec{v}_1^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &\equiv M_{12} = M^x_{12}p_y + M^y_{12}p_y \\
\vec{v}_2^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &= -M_{11} \\
\vec{v}_2^H \cdot H^1_{\vec{K}} \cdot \vec{v}_2 &= (M_{12})^*
\end{aligned} \tag{2}$$

The last two identities are motivated by the effect that $(M)_{ij} = M_{ij}$ must be a Hermitian matrix with no absolute part ($\propto \sigma_0$). To obtain an arbitrary basis of the degnerate subspace I do a $SU(2)$-rotation ($SU(2)$ automatically enforces the orthonormality of the new basis):

$$\begin{pmatrix}\vec{w}_1\\ \vec{w}_2\end{pmatrix} = \begin{pmatrix}e^{i\chi} & 0\\ 0 & e^{i\chi}\end{pmatrix}\begin{pmatrix}\cos(\phi/2)e^{i(\theta/2+\kappa)} & \sin(\phi/2)e^{i\theta/2}\\ \sin(\phi/2)e^{i(-\theta/2+\kappa)} & -\cos(\phi/2)e^{-i\theta/2}\end{pmatrix}\begin{pmatrix}\vec{v}_1\\ \vec{v}_2\end{pmatrix} \tag{3}$$

Parameter ranges?Optimally I would find a Group-theoretical justification of this representaion. χ is a global phase, which I choose to be 0, justified by the $U(1)$-gauge symmetry of quantum mechanics. This leaves three parameters. Now I can write down the projection of $H^1_{\vec{K}}$ onto $\{\vec{w}_1, \vec{w}_2\}$. I write the resulting hermitian 2x2 matrix in the Pauli-matrix basis (again there is no σ_0-part):

$$\begin{aligned}
\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} = \quad & \sigma_x\left[M_{11}\sin(\phi)\cos(\theta) + M_{12}\left(\sin(\kappa)\sin(\theta) - \cos(\phi)\cos(\kappa)\cos(\theta)\right)\right] && (4)\\
& +\sigma_y\left[M_{11}\sin(\phi)\sin(\theta) - M_{12}\left(\sin(\kappa)\cos(\theta) + \cos(\phi)\cos(\kappa)\sin(\theta)\right)\right] && (5)\\
& +\sigma_z\left[M_{11}\cos(\phi) + M_{12}\sin(\phi)\cos(\kappa)\right] && (6)
\end{aligned}$$

One might complain that the equations above dont reference the Fermi velocity. However, as long as we only make statements using ratios of M-components we still obtain the desired result. Now, (ϕ, θ, κ) can be choosen to i.e. give a projected Hamiltonian of the form

$$\langle H^1_{\vec{K}}\rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F\left(p_x\sigma_x + p_y\sigma_y\right). \tag{7}$$

A partial solution is

$$\begin{aligned}
\cos(\phi) &= 0\\
\cos(\kappa) &= 0\\
\tan(\theta) &= -\frac{M^x_{12}}{M^y_{12}}.
\end{aligned} \tag{8}$$

The above is not sufficient to completely fix ϕ and κ as both $\pi/2$ and $-\pi/2$ are still correct solutions, yet only one combination gives the correct result. However, the partial solutions ensures that the σ_z

component vanishes in every case. At the moment I just have to try all 4 possibilities until I found a systematic solution. I expect that the answer lies in the signs of the M_{ij}.

One could (for example) also pick the form

$$\langle H^1_{\vec{K}} \rangle_{\{\vec{w}_1,\vec{w}_2\}} \stackrel{!}{=} v_F \big(p_x \sigma_x + p_y \sigma_z\big). \tag{9}$$

A partial solution for this is

$$\begin{aligned} \tan(\phi) &= \sqrt{-\frac{M^x_{11} M^y_{12}}{M^y_{11} M^x_{12}}} \\ \sin(\kappa) &= 0 \\ \sin(\theta) &= 0. \end{aligned} \tag{10}$$

Again similar problem as above, κ and θ are not completely fixed yet. In what follows I will usually apply the second form as a consistency check.

0.1.2 Straining the tight-binding model

I will only give the relevant equations since I have used this method prior. Straining the tight binding model works as follows:

$$t(\vec{\delta}_{ij}) e^{i\vec{K}\cdot\vec{\delta}_{ij}} \approx t e^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p}\cdot\vec{\delta}_{ij} - \beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) + i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \tag{11}$$

and

$$\begin{aligned} t(\vec{\delta}_{ij}) e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(-\vec{\delta}_{ij}) e^{-i\vec{K}\cdot\vec{\delta}_{ij}} \approx & 2t\cos\left(\vec{q}\cdot\vec{\delta}_{ij}\right) - 2t\sin\left(\vec{q}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\ & -2\beta t\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\ & -2t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \end{aligned} \tag{12}$$

and

$$\begin{aligned} t(\vec{\delta}_{ij}) e^{i\vec{q}\cdot\vec{\delta}_{ij}} - t(-\vec{\delta}_{ij}) e^{-i\vec{q}\cdot\vec{\delta}_{ij}} \approx & 2it\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right) + 2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{p}\cdot\vec{\delta}_{ij}\right) \\ & -2i\beta t\sin\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \\ & +2it\cos\left(\vec{K}\cdot\vec{\delta}_{ij}\right)\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \end{aligned} \tag{13}$$

I will usually split everything into two parts:

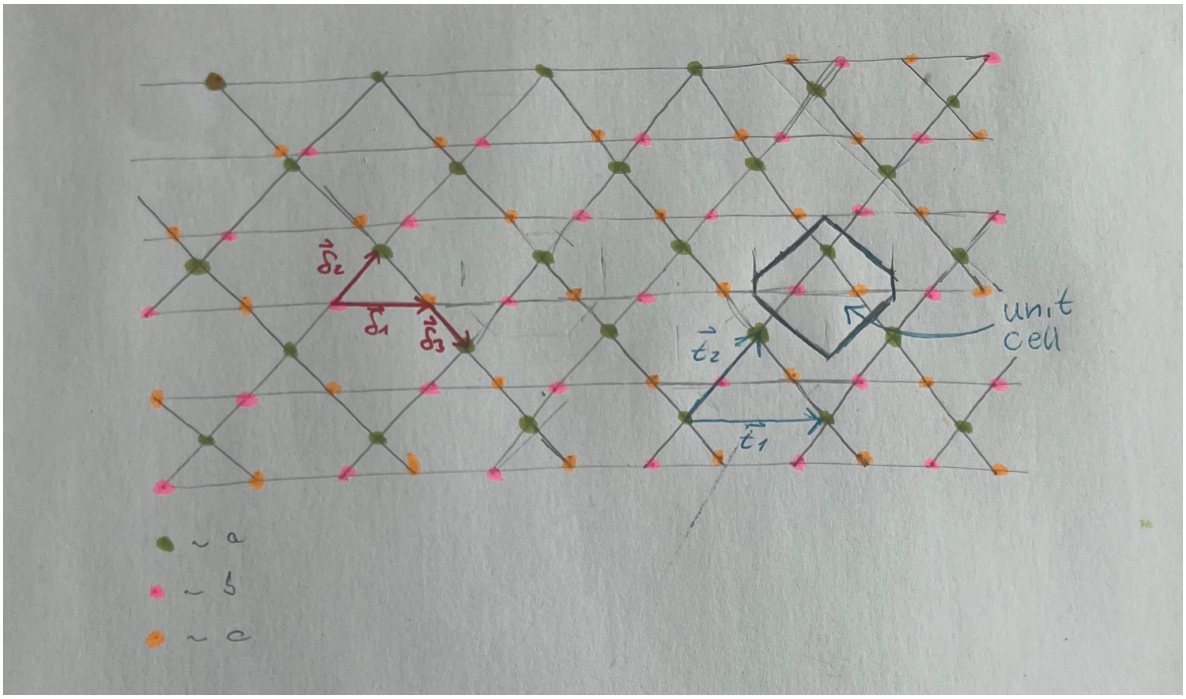
$$\delta H_0 \quad \text{and} \quad \delta H_1, \tag{14}$$

where δH_i is of order i in β.

0.2 0-flux State

0.2.1 Real Space

The bare Kagome lattice is a triangular Bravais lattice with a three site unit cell (called a, b and c-sites below). The Wigner Seitz-cell is a hexagon (with full symmetries, even if the drawing suggests otherwise :)). Geometrically speaking the lattice consists of corner-sharing equilateral triangles.



In all that follows I normalize all distances wrt the distance between nearest neighbours. The nearest neighbour vectors are (see cartoon)

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} \tag{15}$$

and the primitive translations

$$\vec{t}_1 = 2\vec{\delta}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = 2\vec{\delta}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}. \tag{16}$$

The tight binding hamiltonain is

$$H = t\sum_{i\sigma}\left(a^\dagger_{i\sigma}c_{i\sigma} + c^\dagger_{i\sigma}b_{i\sigma} + b^\dagger_{i\sigma}a_{i\sigma} + c^\dagger_{i\sigma}b_{i+t_1;\sigma} + c^\dagger_{i\sigma}a_{i+t_1-t_2;\sigma} + a^\dagger_{i\sigma}b_{i+t_2;\sigma} + \text{h.c.}\right) \tag{17}$$

The first index refers to the unit cell and the second one to the "species" of the particle, the latter being a product of the parton decomposition.

0.2.2 Reciprocal Space

The recipricoal lattice vectors are

$$\vec{b}_1 = \pi\begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = \pi\begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix} \tag{18}$$

The first Brillouin zone is also a hexagon (rotated by 90° wrt to the Wigner-Seitz cell). To find H in reciprocal space I do a Fourier transform:

$$a_{i\sigma} = \frac{1}{\sqrt{N}}\sum_k e^{i\vec{k}\vec{R}_i}a_{k\sigma}, \tag{19}$$

where N is the number of unit cells. This yields

$$H = 2t \sum_{q\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} & c^\dagger_{q\sigma} \end{pmatrix} \begin{pmatrix} 0 & \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & \cos\left(\vec{q}\cdot\vec{\delta}_3\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right) & 0 & \cos\left(\vec{q}\cdot\vec{\delta}_1\right) \\ \cos\left(\vec{q}\cdot\vec{\delta}_3\right) & \cos\left(\vec{q}\cdot\vec{\delta}_1\right) & 0 \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \end{pmatrix}. \tag{20}$$

0.2.3 dispersion and band touching points

The three bands are

$$\varepsilon_1(\vec{q}) = t\left(1 - \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{21}$$

$$\varepsilon_2(\vec{q}) = t\left(1 + \sqrt{4\cos^2(q_x) + 4\cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1}\right) \tag{22}$$

$$\varepsilon_3(\vec{q}) = -2t, \tag{23}$$

see [1]. Interestingly, the top band is completely flat.

Dirac points

The other two bands touch when

$$\cos^2(q_x) + \cos(q_x)\cos\left(\sqrt{3}q_y\right) + 1/4 = 0 \tag{24}$$

$$\Rightarrow \cos(q_x) = \frac{1}{2}\left(-\cos\left(\sqrt{3}q_y\right) \pm \sqrt{\cos^2(\sqrt{3}q_y) - 1}\right) \tag{25}$$

A handy solution to this equation is

$$\begin{aligned} q_x &= (-1)^s \frac{2\pi}{3} + m_1\pi \\ q_y &= \frac{\pi}{\sqrt{3}}\left(2m_2 - m_1\right), \end{aligned} \tag{26}$$

with $s \in 0, 1$ and $m_1, m_2 \in \mathbb{N}$. These touching points lie at the corners of the Brillouin zone. They are all symmetry-equivalent. Representives for the time-revearsal partners in the first Brillouin zone are

$$\vec{K} = \begin{pmatrix} 2\pi/3 \\ 0 \end{pmatrix}, \quad s = 0,\ m_1 = 0,\ m_2 = 0 \tag{27}$$

$$\vec{K}' = \begin{pmatrix} -2\pi/3 \\ 0 \end{pmatrix}, \quad s = 1,\ m_1 = 0,\ m_2 = 0. \tag{28}$$

0.2.4 Linearization in Reciprocal Space and Projection

The linearized Hamiltonian according to

$$H(\vec{K} + \vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}} H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{29}$$

and find

$$H^0_{\vec{K}} = 2t \begin{pmatrix} 0 & \cos(K_2) & \cos(K_3) \\ \cos(K_2) & 0 & \cos(K_1) \\ \cos(K_3) & \cos(K_1) & 0 \end{pmatrix} \tag{30}$$

and

$$H^1_{\vec{K}} = -2t \begin{pmatrix} 0 & \sin(K_2)p_2 & \sin(K_3)p_3 \\ \sin(K_2)p_2 & 0 & \sin(K_1)p_1 \\ \sin(K_3)p_3 & \sin(K_1)p_1 & 0 \end{pmatrix} \tag{31}$$

The eigenvector of $H^0_{\vec{K}}$ corresponding to the flat band reads:

$$\vec{v}_3(\vec{K}) = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}. \tag{32}$$

A possible (orthonormal) basis for the subspace orthogonal to $\vec{v}_3$ is

$$\vec{v}_1(\vec{K}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_2(\vec{K}) = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}. \tag{33}$$

Projeting H onto this basis yields the coefficients

$$\begin{aligned} M^x_{11} &= -\frac{\sqrt{3}}{2}t \\ M^y_{11} &= -\frac{3}{2}t \\ M^x_{12} &= -\frac{3}{2}t \\ M^y_{12} &= \frac{\sqrt{3}}{2}t. \end{aligned} \tag{34}$$

$$\Rightarrow v_F(\vec{K}) = \sqrt{3}t \tag{35}$$

The corresponding parameters for an effective theory of the form $H = \varepsilon_0\sigma_0 + p_x\sigma_x p_y\sigma_y$ are

$$\begin{aligned} \phi &= -\frac{\pi}{2} \\ \kappa &= -\frac{\pi}{2} \\ \theta &= \arctan\Big(\sqrt{3}\Big) = \frac{\pi}{3}. \end{aligned} \tag{36}$$

Introducing strain yields the following extra expansion terms of H:

$$\delta H_1 = -2t\beta \begin{pmatrix} 0 & (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy}/8 \\ (u_{xx} + 2\sqrt{3}u_{yx} + 3u_{yy})/8 & 0 & -u_{xx}/2 \\ (u_{xx} - 2\sqrt{3}u_{yx} + 3u_{yy}/8 & -u_{xx}/2 & 0 \end{pmatrix} \tag{37}$$

$$\delta H_2 = -2t \begin{pmatrix} 0 & \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 \\ \pi(\frac{u_{xx}}{\sqrt{3}} + u_{yx})/2 & 0 & \pi\frac{u_{xx}}{\sqrt{3}} \\ \pi(\frac{u_{xx}}{\sqrt{3}} - u_{yx})/2 & \pi\frac{u_{xx}}{\sqrt{3}} & 0 \end{pmatrix}. \tag{38}$$

Projecting these onto the correct basis gives

$$\langle \delta H_1 \rangle_{\{\vec{w}_1,\vec{w}_2\}} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{3}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{3}\frac{\beta}{\sqrt{3}}u_{yx} \tag{39}$$

$$\langle \delta H_0 \rangle_{\{\vec{w}_1,\vec{w}_2\}} = \sqrt{3}\frac{2\pi}{3}u_{xx}\sigma_x + \sqrt{3}\frac{2\pi}{3}u_{yx}. \tag{40}$$

Thus, the final result is

$$\phi(\vec{K}) = -\frac{\beta}{2}(u_{xx}+u_{yy}) \tag{41}$$

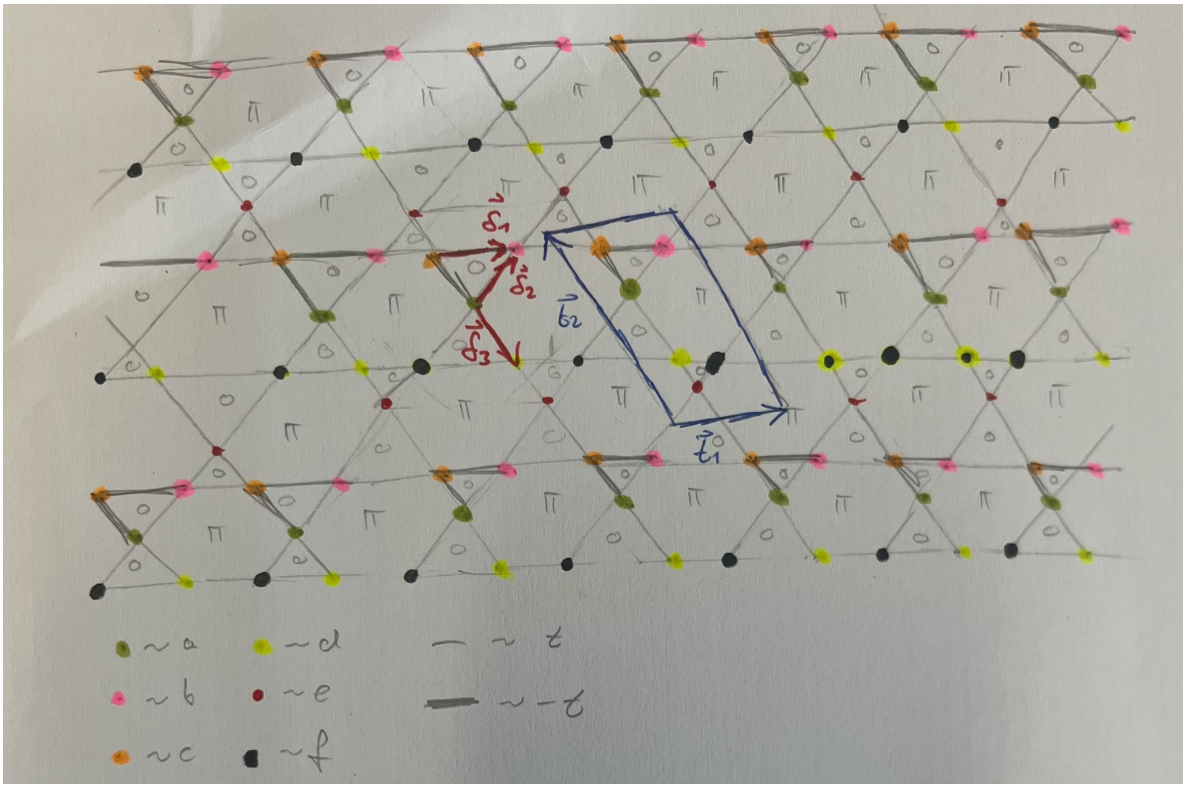
$$\vec{A}(\vec{K}) = \frac{\beta}{2\sqrt{3}}\begin{pmatrix} u_{yy}-u_{xx} \\ 2u_{yx} \end{pmatrix} \tag{42}$$

$$\Rightarrow \vec{B}(\vec{K}) = -\frac{\beta}{2\sqrt{3}}(\partial_x\partial_y U_y - \partial_x^2 U_x) \tag{43}$$

0.3 π-flux State

0.3.1 Real Space

This system is a rectengular Bravais-lattice with a six site unit cell. It emerges from the bare Kagome lattice through unit cell doubling. Even though the Wigner-Seitz-cell is a rectangle I will work with the unit cell shown below.



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2}\begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \tag{44}$$

The new primitve translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \tag{45}$$

The real space (effective) tight-binding hamiltonian is

$$\begin{aligned} H = t \sum_{i\sigma} \Big(& - a^\dagger_{i\sigma} c_{i\sigma} - c^\dagger_{i\sigma} b_{i\sigma} + b^\dagger_{i\sigma} a_{i\sigma} + a^\dagger_{i\sigma} d_{i\sigma} + d^\dagger_{i\sigma} e_{i\sigma} + e^\dagger_{i\sigma} f_{i\sigma} + f^\dagger_{i\sigma} d_{i\sigma} \\ & + b^\dagger_{i\sigma} c_{i+t_1;\sigma} + b^\dagger_{i\sigma} e_{i+t_1+t_2;\sigma} + f^\dagger_{i\sigma} d_{i+t_1;\sigma} + f^\dagger_{i\sigma} a_{i+t_1;\sigma} + e^\dagger_{i\sigma} c_{i-t_2;\sigma} + \text{h.c.} \Big) \end{aligned} \tag{46}$$

0.3.2 Reciprocal Space and Bands

The reciprocal lattice vectrors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \tag{47}$$

I do a Fouriertransform

$$\Psi_{q\sigma} \equiv \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \\ c_{q\sigma} \\ e_{q\sigma} \\ f_{q\sigma} \\ d_{q\sigma} \end{pmatrix} = \frac{1}{\sqrt{N}} \sum_i e^{i\vec{q}\vec{R}_i} \begin{pmatrix} a_{i\sigma} \\ b_{i\sigma} \\ c_{i\sigma} \\ e_{i\sigma} \\ f_{i\sigma} \\ d_{i\sigma} \end{pmatrix} \tag{48}$$

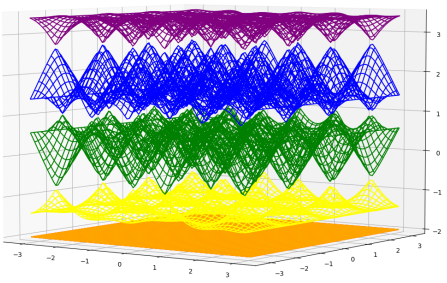
(N is the numer of unit cells.) The definition of Ψ is a bit awkward. If I write this down somewhere serious I should just label the sites differently.

$$H = t \sum_{q,\sigma} \Psi^\dagger_{q\sigma} \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i\sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{iq_3} & -2i\sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2\cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2\cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \tag{49}$$

where the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$ was used.

Bands

The dispersion relation is shown in the following picture



This plot was generatey by numerically calculating the eigenvalues of H on a (q_x, q_y)-grid. The lowest band (flat band) is doubly degenerate.

The characteristic polynomial of 49 is (this was generate by mathematica)

$$x^6 - 12x^4 - 8x^3 + x^2(30 + 2\Delta_{\vec{q}}) + x(24 + 8\Delta_{\vec{q}}) + 8\Delta_{\vec{q}} - 8 \tag{50}$$

units! with

$$\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{51}$$

I should verify this uisng Cayley-Hamilton. To derive analytical expressions for the bands I first divide out the two flat bands (i.e. polynomial division by $(x+2)^2$). The result is

$$x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}}. \tag{52}$$

To make progress I postulate (judging from the numeriacl plot of the dispersion) that the above polynomial can be written as a product of two quadratic polynomials. Furtheremore, the solutions of said qaudratic equations must be centered around t. The p-q forumla suggests that the linear coefficents of both quadratic polynomials must therefore be $-2t$.

$$\begin{aligned} \Rightarrow x^4 - 4x^3 + 8x - 2 + 2\Delta_{\vec{q}} &= (x^2 - 2x + a) \cdot (x^2 - 2x + b) \\ &= x^4 - 4x^3 + x^2(4 + a + b) - 2x(a + b) + ab. \end{aligned} \tag{53}$$

$$\Rightarrow a + b \stackrel{!}{=} -4, \quad ab \stackrel{!}{=} -2 + 2\Delta_{\vec{q}}. \tag{54}$$

These equations are solved by

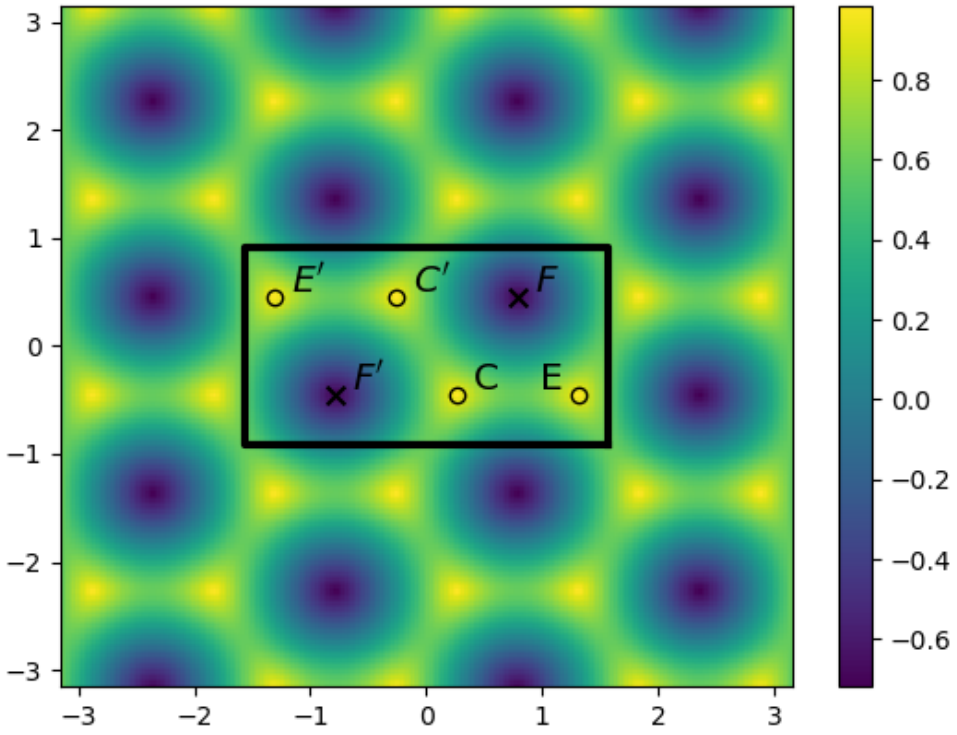
$$\begin{aligned} a_{1,2} &= -2 \pm \sqrt{6 - 2\Delta_{\vec{q}}} \\ b_{1,2} &= -2 \mp \sqrt{6 - 2\Delta_{\vec{q}}}. \end{aligned} \tag{55}$$

Both solution lead to the same result (a and b are exchanged in the second soltuion wrt the first one). This gives the following expressions for the bands

$$
\begin{aligned}
\varepsilon_1(\vec{q}) &= -2t \\
\varepsilon_2(\vec{q}) &= -2t \\
\varepsilon_3(\vec{q}) &= \left(1 - \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_4(\vec{q}) &= \left(1 - \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_5(\vec{q}) &= \left(1 + \sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t \\
\varepsilon_6(\vec{q}) &= \left(1 + \sqrt{3 + \sqrt{6 - 2\Delta_{\vec{q}}}}\right) t.
\end{aligned} \tag{56}
$$

$$
\Delta_{\vec{q}} = -\cos(4q_2) + \cos(4q_3) - \cos(4q_1). \tag{57}
$$

Band Touching Points



Shown above is the 4th band (green band). The black rectangle is the First Brillouin zone. Also shown are the position of the various band-touching points. F the Dirac points at 1/2- (and 5/6-) filling. At 2/3-filling the Dirac points are at C and E.

Now I derive the coordiantes. It is clear from the expressions above that bands 3 and 4 as well as bands 5 and 6 touch iff

$$
\begin{aligned}
&6 - 2\Delta_{\vec{q}} = 0 \\
\Leftrightarrow &-\cos(4q_2) + \cos(4q_3) - \cos(4q_1) = 3 \\
\Leftrightarrow &\, 2\sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) - \cos(4q_x) = 3.
\end{aligned} \tag{58}
$$

This can easily be solved considering the ranges of sine and cosine. A possible parametrization of the solution is

$$q_x = (-1)^s \frac{\pi}{4} + m_1\pi \tag{59}$$

$$q_y = (-1)^s \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}}, \tag{60}$$

where

$$s \in [0, 1], \quad m_1, m_2 \in \mathbb{Z}.$$

Both $(s = 0, m_1 = 0, m_2 = 0)$ and $(s = 1, m_1 = 0, m_2 = 0)$ are in the first Brillouin zone:

$$\vec{F} = \frac{\pi}{4}\begin{pmatrix}1\\ 1/\sqrt{3}\end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4}\begin{pmatrix}1\\ 1/\sqrt{3}\end{pmatrix} \tag{61}$$

.
On the other hand, bands 4 and 5 touch when

$$
\begin{aligned}
&\sqrt{3 - \sqrt{6 - 2\Delta_{\vec{q}}}} = 0 \\
\Leftrightarrow &\, 2\Delta_{\vec{q}} = -3 \\
\Leftrightarrow &\, 4\sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) - 2\cos(4q_x) = -3
\end{aligned} \tag{62}
$$

Using $\cos(2x) = 1 - 2\sin^2(x)$ I obtain

$$
\begin{aligned}
&\sin^2(2q_x) + \sin(2q_x)\sin\Big(2\sqrt{3}q_y\Big) + \frac{1}{4} = 0 \\
\Rightarrow &\sin(2q_x) = \frac{1}{2}\left(-\sin\Big(2\sqrt{3}q_y\Big) \pm \sqrt{\sin^2(2\sqrt{3}q_y) - 1}\right)
\end{aligned} \tag{63}
$$

The expression under the square root gives a condition for q_y. It tourns out there are two distinct sets of solutions (meaning there are two distinct sets of symmetry-equivalent touching points).

$$q_x = (-1)^s \frac{\pi}{12} + m_1\pi \tag{64}$$

$$q_y = (-1)^{s+1} \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}} \tag{65}$$

and

$$q_x = (-1)^s \frac{5\pi}{12} + m_1\pi \tag{66}$$

$$q_y = (-1)^{s+1} \frac{\pi}{4\sqrt{3}} + (m_1 + m_2)\frac{\pi}{\sqrt{3}}, \tag{67}$$

with the usual parmeters. Once more, there are two of each in the first Brillouin zone corresponding to $(s = 0, m_1 = 0, m_2 = 0)$ and $(s = 1, m_1 = 0, m_2 = 0)$:

x

$$\vec{C} = \frac{\pi}{12}\begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}, \quad \vec{C}' = \frac{\pi}{12}\begin{pmatrix}-1\\\sqrt{3}\end{pmatrix},$$
$$\vec{E} = \frac{\pi}{12}\begin{pmatrix}5\\-\sqrt{3}\end{pmatrix}, \quad \vec{E}' = \frac{\pi}{12}\begin{pmatrix}-5\\\sqrt{3}\end{pmatrix} \tag{68}$$

Linearization of H/ Field Theory

To derive an effective (linear) theory in the vicinity of the touching points I linearize the Hamiltonian in $\vec{q}$.

$$H(\vec{K}+\vec{p}) \approx \underbrace{H(\vec{K})}_{\equiv H^0_{\vec{K}}} + \underbrace{\vec{p}\cdot\left(\vec{\nabla}_{\vec{q}}H(\vec{q})\right)\Big|_{\vec{q}=\vec{K}}}_{\equiv H^1_{\vec{K}}} + \mathcal{O}(p^2). \tag{69}$$

$$H^0_{\vec{K}} = \begin{pmatrix} 0 & e^{iK_2} & -e^{-iK_3} & 0 & e^{-iK_2} & e^{iK_3} \\ e^{-iK_2} & 0 & 2i\sin(K_1) & e^{iK_2} & 0 & 0 \\ -e^{iK_3} & -2i\sin(K_1) & 0 & e^{-iK_3} & 0 & 0 \\ 0 & e^{-iK_2} & e^{iK_3} & 0 & e^{iK_2} & e^{-iK_3} \\ e^{iK_2} & 0 & 0 & e^{-iK_2} & 0 & 2\cos(K_1) \\ e^{-iK_3} & 0 & 0 & e^{iK_3} & 2\cos(K_1) & 0 \end{pmatrix} \tag{70}$$

$$H^1_{\vec{K}} = \begin{pmatrix} 0 & ie^{iK_2}p_2 & ie^{-iK_3}p_3 & 0 & -ie^{-iK_2}p_2 & ie^{iK_3}p_3 \\ -ie^{-iK_2}p_2 & 0 & 2i\cos(p_1) & ie^{iK_2}p_2 & 0 & 0 \\ -ie^{iK_3}p_3 & -2i\cos(p_1) & 0 & -ie^{-iK_3}p_3 & 0 & 0 \\ 0 & -ie^{-iK_2}p_2 & ie^{iK_3}p_3 & 0 & ie^{iK_2}p_2 & -ie^{-iK_3}p_3 \\ ie^{iK_2}p_2 & 0 & 0 & -ie^{-iK_2}p_2 & 0 & -2\sin(p_1) \\ -ie^{-iK_3}p_3 & 0 & 0 & ie^{iK_3}p_3 & -2\cos(p_1) & 0 \end{pmatrix} \tag{71}$$

$$K_i \equiv \vec{K}\cdot\vec{\delta}_i$$
$$p_i \equiv \vec{p}\cdot\vec{\delta}_i$$

0.3.3 1/2-Filling

The Dirac points are at

$$\vec{F} = \frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4}\begin{pmatrix}1\\1/\sqrt{3}\end{pmatrix}. \tag{72}$$

I focus on $\vec{F}$ for now. At $\vec{F}$ I choose the basis

$$\vec{v}_3(\vec{F}) = \frac{1}{2\sqrt{6-3\sqrt{3}}}\left(\tfrac{1-\sqrt{3}}{2} + \tfrac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \tfrac{1-\sqrt{3}}{2} + \tfrac{\sqrt{3}-3}{2}i, \quad 0, \quad 1\right)^T \tag{73}$$

$$\vec{v}_4(\vec{F}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}}\left((3-2\sqrt{3}) + (-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3}+(2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3}\right)^T \tag{74}$$

I obtain

$$
\begin{aligned}
M_{33}^{x} &= \frac{1-\sqrt{3}}{2}t \\
M_{33}^{y} &= -\frac{1+\sqrt{3}}{2}t \\
M_{34}^{x} &= -\frac{1+\sqrt{3}}{2}t \\
M_{34}^{y} &= -\frac{1-\sqrt{3}}{2}t
\end{aligned}
\tag{75}
$$

$$
\Rightarrow v_{\vec{F}}^{\frac{1}{2}}(\vec{F}) = \sqrt{2}t \tag{76}
$$

The correct parameters for a theory of the form $H = \epsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ are

$$
\begin{aligned}
\phi &= -\frac{\pi}{2} \\
\kappa &= -\frac{\pi}{2} \\
\theta &= \arctan\Big(2+\sqrt{3}\Big) = \frac{5\pi}{12}
\end{aligned}
\tag{77}
$$

The projections onto these "correct" basis vectors are

$$
\begin{aligned}
\langle \delta H_1 \rangle_{\vec{w}_1,\vec{w}_2} &= -\frac{\beta}{2}(1-\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \\
\langle \delta H_0 \rangle_{\vec{w}_1,\vec{w}_2} &= \sqrt{2}\frac{\pi}{4}\left(u_{xx}+\frac{u_{yx}}{\sqrt{3}}\right)\sigma_x + \sqrt{2}\frac{\pi}{4}\left(u_{xy}+\frac{u_{yy}}{\sqrt{3}}\right)\sigma_y.
\end{aligned}
\tag{78}
$$

Hence, there is no pseudomagnetic-field but a scalar potential:

$$
\phi^{\frac{1}{2}}(\vec{F}) = -\frac{\beta}{2}(1-\sqrt{3})(u_{xx}+u_{yy})\sigma_0 \tag{79}
$$

$$
\vec{A}^{\frac{1}{2}}(\vec{F}) = 0. \tag{80}
$$

0.3.4 2/3-Filling

The Dirac points are at

$$
\begin{aligned}
\vec{C} &= \frac{\pi}{12}\begin{pmatrix}1\\-\sqrt{3}\end{pmatrix}, \quad \vec{C}' = \frac{\pi}{12}\begin{pmatrix}-1\\\sqrt{3}\end{pmatrix}, \\
\vec{E} &= \frac{\pi}{12}\begin{pmatrix}5\\-\sqrt{3}\end{pmatrix}, \quad \vec{E}' = \frac{\pi}{12}\begin{pmatrix}-5\\\sqrt{3}\end{pmatrix}.
\end{aligned}
\tag{81}
$$

I will only consider $\vec{E}$ and $\vec{C}$.

$\tilde{\mathbf{C}}$

At $\vec{C}$ I choose the vectors

$$\vec{v}_4(\vec{C}) = \frac{1}{\sqrt{6}\sqrt{5\sqrt{3}+9}}\left(\frac{1+\sqrt{3}}{\sqrt{2}} - i\sqrt{2}(2+\sqrt{3}) \quad 2+\sqrt{3} \quad i\sqrt{2}(2+\sqrt{3}) \quad \frac{1+\sqrt{3}}{\sqrt{2}} + i\sqrt{2}(2+\sqrt{3}) \quad 1 \quad 0\right) \tag{82}$$

$$\vec{v}_5(\vec{C}) = \frac{\sqrt{2}}{\sqrt{3\sqrt{3}(1+\sqrt{3})}}\left(-\frac{1}{2} \quad -\frac{3}{2\sqrt{2}}(1+\sqrt{3}) \quad i\frac{1+\sqrt{3}}{2} \quad -\frac{1}{2} \quad \frac{1+\sqrt{3}}{2\sqrt{2}} \quad 1\right) \tag{83}$$

.
The matrix elements are

$$\begin{aligned} M^x_{44} &= -\frac{1+\sqrt{3}}{2}t \\ M^y_{44} &= -\frac{1-\sqrt{3}}{2}t \\ M^x_{45} &= -\frac{1-\sqrt{3}}{2}t \\ M^y_{45} &= \frac{1+\sqrt{3}}{2}t \end{aligned} \tag{84}$$

$$\Rightarrow v_F^{\frac{2}{3}}(\vec{C}) = \sqrt{2}t \tag{85}$$

The correct parameters for a theory of the form $\varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_y$ are

$$\begin{aligned} \phi &= -\frac{\pi}{2} \\ \kappa &= -\frac{\pi}{2} \\ \theta &= -\arctan\left(2-\sqrt{3}\right) = -\frac{\pi}{12}. \end{aligned} \tag{86}$$

The resulting projections are

$$\langle \delta H_1 \rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{xx}+u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_y \tag{87}$$

$$\langle \delta H_0 \rangle_{\vec{w}_4,\vec{w}_5} = \sqrt{2}\frac{\pi}{12}(u_{xx}-\sqrt{3}u_{yx})\sigma_x + \sqrt{2}\frac{\pi}{12}(u_{xy}-\sqrt{3}u_{yy})\sigma_y \tag{88}$$

For verification pourposes I will also consider a second theory of the form $\varepsilon_0\sigma_0 + p_x\sigma_x + p_y\sigma_z$, the corresponding parameters are

$$\begin{aligned} \phi &= \arctan\left(7+4\sqrt{3}\right) = \frac{5\pi}{12} \\ \kappa &= 0 \\ \theta &= \pi. \end{aligned} \tag{89}$$

In this case I find the projections

$$\langle \delta H_1 \rangle_{\vec{w}_4,\vec{w}_5} = -\frac{\beta}{2}(u_{yy}+u_{xx})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{yy}-u_{xx})\sigma_x + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy}+u_{yx})\sigma_z \tag{90}$$

$$\langle \delta H_0 \rangle_{\vec{w}_4,\vec{w}_5} = ? \tag{91}$$

The two cases agree on

$$\phi^{\frac{2}{3}}(\vec{C}) = -\frac{\beta}{2}(u_{xx} + u_{yy}) \tag{92}$$

$$\vec{A}^{\frac{2}{3}}(\vec{C}) = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{yy} - u_{xx} \\ u_{xy} + u_{yx} \end{pmatrix}. \tag{93}$$

$$\Rightarrow B^{\frac{2}{3}}(\vec{C}) = \frac{\beta}{2\sqrt{3}} \left(2\partial_x \partial_y U_x + \left[\partial_x^2 - \partial_y^2\right] U_y\right) \tag{94}$$

.

$\tilde{\mathbf{E}}$

At $\vec{E}$ I choose the basis vectors

$$\vec{v}_4(\vec{E}) = \frac{1}{\sqrt{6(9-5\sqrt{3})}} \left(\sqrt{2}(2-\sqrt{3}) + i\frac{1-\sqrt{3}}{\sqrt{2}} \quad -2+\sqrt{3} \quad i\sqrt{2}(2-\sqrt{3}) \quad \sqrt{2}(2-\sqrt{3}) - i\frac{1-\sqrt{3}}{\sqrt{2}} \quad 1 \quad 0\right) \tag{95}$$

$$\vec{v}_5(\vec{E}) = \frac{\sqrt{2}}{\sqrt{3(3-\sqrt{3})}} \left(\frac{i}{2} \quad \frac{\sqrt{3}}{2\sqrt{2}}(1-\sqrt{3}) \quad -i\frac{1-\sqrt{3}}{2} \quad -\frac{i}{2} \quad -\frac{1-\sqrt{3}}{2\sqrt{2}} \quad 1\right). \tag{96}$$

The projection coefficents are

$$\begin{aligned} M_{44}^x &= -\frac{1-\sqrt{3}}{2}t \\ M_{44}^y &= \frac{1+\sqrt{3}}{2}t \\ M_{45}^x &= -\frac{1+\sqrt{3}}{2}t \\ M_{45}^y &= -\frac{1-\sqrt{3}}{2}t. \end{aligned} \tag{97}$$

$$\Rightarrow v_F^{\frac{2}{3}}(\vec{E}) = \sqrt{2}t \tag{98}$$

The correct parameters are

$$\begin{aligned} \phi &= \pi/2 \\ \kappa &= -\pi/2 \\ \theta &= \arctan\left(2+\sqrt{3}\right) = \frac{5\pi}{12}. \end{aligned} \tag{99}$$

The cooresponding projections are

$$\langle \delta H_1 \rangle_{\vec{w}_4, \vec{w}_5} = -\frac{\beta}{2}(u_{xx} + u_{yy})\sigma_0 + \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xx} - u_{yy})\sigma_x - \sqrt{2}\frac{\beta}{2\sqrt{3}}(u_{xy} + u_{yx})\sigma_y \tag{100}$$

$$\langle \delta H_0 \rangle_{\vec{w}_4, \vec{w}_5} = \sqrt{2}\frac{\pi}{12}\left(5u_{xx} - \sqrt{3}u_{yx}\right)\sigma_x + \sqrt{2}\frac{\pi}{12}(5u_{xy} - \sqrt{3}u_{yy})\sigma_y. \tag{101}$$

Therefore,

$$\phi^{\frac{2}{3}}(\vec{E}) = -\frac{\beta}{2}(u_{xx} + u_{yy}) \tag{102}$$

$$\vec{A}^{\frac{2}{3}}(\vec{E}) = \frac{\beta}{2\sqrt{3}} \begin{pmatrix} u_{xx} - u_{yy} \\ -(u_{xy} + u_{yx}) \end{pmatrix} \tag{103}$$

It turns out $\vec{A}^{\frac{2}{3}}(\vec{E}) = -\vec{A}^{\frac{2}{3}}(\vec{C})$.

I verified tthis result by doing the calculation for $\vec{E} + \vec{b}_2$ and obtaining the same results. However, the eigenvectors were not the same why?

0.3.5 5/6-filling

The Dirac points are once more at

$$\vec{F} = \frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{F}' = -\frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \tag{104}$$

I will do the calculations for $\vec{F}$. I choose the basis vecors

$$\vec{v}_5(\vec{F}) = \frac{1}{\sqrt{\sqrt{6} + 3\sqrt{2}}} \left(-\tfrac{1}{\sqrt{2}} - i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad -2 - \sqrt{3} \quad i\sqrt{2}(1+\sqrt{3}) \quad -\tfrac{1}{\sqrt{2}} + i\tfrac{2+\sqrt{3}}{\sqrt{2}} \quad 1 \quad 0 \right) \tag{105}$$

$$\vec{v}_6(\vec{F}) = \frac{1}{4\sqrt{3}} \left(1 + \sqrt{3} + i(1-\sqrt{3}) \quad \sqrt{2} + \sqrt{6} \quad 0 \quad 1 + \sqrt{3} - i(1-\sqrt{3}) \quad \sqrt{2} + \sqrt{6} \quad 4 \right) \tag{106}$$

The matrix elements are

$$\begin{aligned} M^x_{55} &= \frac{1+\sqrt{3}}{2} t \\ M^y_{55} &= -\frac{1-\sqrt{3}}{2} t \\ M^x_{56} &= \frac{1-\sqrt{3}}{2} t \\ M^y_{56} &= \frac{1+\sqrt{3}}{2} t. \end{aligned} \tag{107}$$

$$\Rightarrow v_F^{\frac{5}{6}}(\vec{F}) = \sqrt{2}t \tag{108}$$

Hence, the correct parameters are

$$\begin{aligned} \phi &= \pi/2 \\ \kappa &= -\pi/2 \\ \theta &= \arctan\left(2 - \sqrt{3}\right) = \frac{\pi}{12} \end{aligned} \tag{109}$$

Applying strain yields the results

$$\langle \delta H_1 \rangle_{\vec{w}_5, \vec{w}_6} = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx} + u_{yy})\sigma_0 \tag{110}$$

$$\langle \delta H_0 \rangle_{\vec{w}_5, \vec{w}_6} = \sqrt{2}\frac{\pi}{4} \left(u_{xx} + \frac{u_{yx}}{\sqrt{3}} \right) \sigma_x + \sqrt{2}\frac{\pi}{4} \left(u_{xy} + \frac{u_{yy}}{\sqrt{3}} \right) \sigma_y \tag{111}$$

$$\phi^{\frac{5}{6}}(\vec{F}) = -\frac{\beta}{2}(1+\sqrt{3})(u_{xx} + u_{yy})\sigma_0 \tag{112}$$

$$\vec{A}^{\frac{5}{6}}(\vec{F}) = 0. \tag{113}$$

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